LP Modeling

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Lookup

Runtime

minimize: nicht linear!

$$
\begin{aligned}
t_R^{L_1} \quad & +p_K^{L_1} * t_l^{L_1} \\
& +(1-p_K^{L_1}) * \Bigg(t_R^{L_2} \quad +p_K^{L_2} * t_l^{L_2} \\
& \qquad +(1-p_K^{L_2}) * \Big(t_R^{L_3} \quad +p_K^{L_3} * t_l^{L_3} \\
& \qquad\qquad +(1-p_K^{L_3}) * t^H \Big) \Bigg)
\end{aligned}
$$

TODO: linearise

variable	meaning
$p_K^{L_i}$	probability of a kmer belonging to level i
$t_R^{L_i}$	time to check presence of a bit in vector R of level i
$t_l^{L_i}$	time to lookup a k-mer on level i

for: **eines muss ne summe sein!**

$$p_K^{L_i} = \prod_{m \in M} \left(sm_m^{L_i} \prod_{t \in T} st_t^{L_i} * p_{m-empiric}^{L_i}(m,t) \right)$$

TODO: linearise

variable	meaning
$sm_m^{L_i}$	selector for minimizer length m in level i
$st_t^{L_i}$	selector for threshold of t in level i
$p_{m-empiric}^{L_i}(m,t)$	empirically estimated value for a kmer with a minimizer of length m to have a frequency less than t
M	Set of all minimizer length to be considered
T	Set of thresholds to be considered

selectors for m and t must be mutually exclusive:

$$\sum_{m \in M} sm_m^{L_i} = 1$$

$$\sum_{t \in T} st_t^{L_i} = 1$$

Space Consumption

Using Elias Fano Coding a Bitvector of length U with n bits set to 1 can be encoded using $n\lceil log\frac{U}{n}\rceil + 3n$.

Let $noam$ be the number of appearing minimizers (minimizers with a frequency greater than zero).

Level 1

R - Presence of k-mers:
probably not relevant R^{L_1} has length $4^{sm_m^{L_1}}$ and contains $p_M^{L_1} * noam$ many 1s.

$$\begin{aligned} s_R^{L_1} &= p_M^{L_1} * noam\lceil log\frac{4^{sm_m^{L_1}}}{p_M^{L_1} * noam}\rceil + 3 * p_M^{L_1} * noam \\ &= p_M^{L_1} * noam\left(\lceil log\frac{4^{sm_m^{L_1}}}{p_M^{L_1} * noam}\rceil + 3\right) \end{aligned}$$

S - Offset Boundaries:
S is a bitvector of length $|O| + 1$ and is **probably not relevant** for the optimization of m and t.

O - Offsets:
O^{L_1} contains one entry of size $log|T|$ for the offset of every minimizer with frequency $\leq t^{L_1}$ in T.

$$|O| = log_2|T| * \sum_{m=1}^{t^{L_1}} m * (\text{number of minimizers with frequenzy m})$$

Every minimizer of length m covers an average of $\frac{k-m+2}{2}$ consecutive k-mers.

$$bpk_O^{L_1} = \frac{2}{k - m^{L_1} + 2} * p_K^{L_1} log|T|$$